

Semiconductor Research Corporation: A Case Study in Cooperative Innovation Partnerships

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Abstract In the study of innovation institutions, it is important to consider how different institutional models can affect a research organization in conducting or funding successful work. As an industry collaborative, Semiconductor Research Corporation (SRC) provides an example of a privately funded institution that leverages the inputs of several member companies, along with federal funding, to accomplish innovation in its mission area. SRC has several component programs, all attempting to find innovative solutions to semiconductor problems, but on different time scales, and in different technology areas. But how does SRC use its resources to ensure these goals? Through data gathered from semi-structured qualitative interviews and SRC documentation, this paper addresses that question. SRC has found a way to leverage industry money to motivate and develop a robust field of university research for over 30 years. SRC uses several mechanisms for maintaining an application focused, member-centered decision process, institutional flexibility, and strong ties between industry contributors and university researchers. SRC has continued to keep its members satisfied by training thousands of graduate students for employment in their member companies, by focusing on precompetitive research that addresses industry requirements, and doing so in a manner that operates leanly, with low overhead to its funders. Given these successes, we identify aspects of SRC operations, such as a focus on its member company needs, frequent interactions between funders and researchers, flexible funding mechanisms, and focus on workforce development, that may be diffusible to innovation institutions, including federal research efforts.

Keywords Technology policy · Innovation · Energy · Semiconductors · Institutions

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Introduction

A significant amount of federal spending has gone to research and development in the United States. Although only a small percentage of overall spending, the U.S. government has spent \$50 trillion on research and development from 1950 to 1999 (Apte and Scalise 2009). During this time, industrial research in pharmaceuticals, defense applications, electronics, energy, and consumer products has also provided significant amounts of research funding, which resulted in an abundance of useful knowledge and inventions. Given the large amounts spent on research, and the great potential for wasted effort in highly uncertain scientific fields, more scrutiny is necessary to understand both how the policies that shape research efforts work, and how institutions that fund research can be both efficient and effective.

This study on the Semiconductor Research Corporation (SRC) is part of a larger program of research on innovation institutions. This larger effort aims to understand energy technology innovation institutions, their management of science and technology research prioritization and implementation, and how their organizational structure can affect outcomes. In studying one institution, the National Renewable Energy Laboratory (NREL), we explored a federal, mission-oriented laboratory focused in one area of application, renewable energy innovation. SRC provided a comparison case. The semiconductor industry created SRC as a direct response to perceived shortfalls in the future capabilities of the U.S. semiconductor industry, especially in the face of international competition from Japan. In addition to competition, energy innovation faces obstacles due to existing infrastructure, political considerations, and environmental concerns. Additionally, there are technological overlaps between the photovoltaic (PV) area of energy innovation and semiconductors. Despite the similarities between the semiconductor products and PV development, SRC represents a greatly different funding model from many federal institutions, universities, and corporate laboratories. Unlike the energy field, the U.S. semiconductor industry has come together in a cooperative partnership to collaboratively fund programs to address perceived shortfalls in innovation and workforce development. SRC sees its role as facilitating partnerships between industry partners and academics, and leveraging both industry and government funding in an attempt to utilize university research inquiry for industry-specific applications.

Much of industry research in the 1950s, 1960s, and the 1970s took place inside large corporate labs. Large, vertically integrated companies like Bell Labs and IBM had the capacity to conduct sizable basic research programs and to link such programs with technology development and manufacturing activities. However, in the 1970s, many semiconductor companies were not able to devote sufficient resources to funding long-term research and became less capable of keeping up with competitors and of ensuring future technological development. According to one source, “SRC exists because market competition forced increases in company resources being used for short-range product development with a decreasing share for long-range research. The industry created SRC in order to merge these decreasing shares with those of other companies to sustain an effective and essential research activity” (Burger 2000).

Additionally, many industry leaders perceived a lack of skilled talent in the workforce, and international competition loomed as a burgeoning threat. In the late 1970s and early 1980s, the Japanese rise in consumer electronics caused pressure on the semiconductor industry. In 1981, six of the top ten semiconductor producers were from the U.S.; the other four were Japanese companies. However, Japanese companies had captured 42% of the market for 16k DRAMS¹ in 1980 (Burger 2000), and 70% in 1981 (*Fortune* 1981). Over the 8 years from 1978 to 1986, the Japanese share of the market grew from 30% to 75%, while the U.S. share shrunk from 70% to 20% (Economist 1980).

Robert Noyce, co-founder of Intel and co-creator of the integrated circuit, announced the creation of SRC in 1981. SRC would be an independent subsidiary of the Semiconductor Industry Association (SIA), existing to increase the participation of university academics in the development of semiconductor technology. He attributed its creation to a need “to channel more funds to research,” along with a need to “add to the supply and quality of degreed professional people” (SIA 1982) (Etzkowitz 1984). With success defined as driving increased university attention to semiconductor-relevant research, and augmenting the industry’s workforce by training new professionals, SRC has been successful. In 2011, SRC programs trained almost 1,500 graduate students, which is almost 15 times the number of students doing industry-relevant research in the U.S. in 1982, with these research programs also contributing publications and patents in the field.

A group of paying industry members funded SRC to invest in relevant academic research, intending to help their companies compete in the international semiconductor market. This represented a new type of arrangement for shared research effort between otherwise competitive companies. Over time, SRC has expanded from its original research program, which is now the Global Research Collaboration (GRC), to include two additional major initiatives also focused on semiconductor-related research, the Focus Center Research Program (FCRP) and the Nanoelectronics Research Initiative (NRI). SRC also has the capability to establish new topical research programs, such as the Energy Research Initiative (ERI). Each program is a legally distinct entity that is a wholly-owned subsidiary of SRC. The programs address different time frames and technology areas, and thus also differ somewhat in industry membership, structure, and management.

We explore and discuss the SRC model and how it operates. We attempt to draw out lessons from SRC for possible dissemination and application at other institutions, especially those attempting to deliver useful energy innovation. The first lesson largely hinges on whether the SRC model is successful for its own needs, and the second on whether the SRC model is successful when compared to other institutional models.

SRC is largely a privately funded endeavor. With the leverage of some government funding, it has utilized industry money to promote the development of a strong semiconductor research system. This has worked to its and the industry’s benefit. Since its inception, SRC has slowly grown to encompass a greater range of scientific problems and new types of research, while continually funding the

¹ Dynamic Random-Access Memory, which supplies memory to electronic devices.

medium-term research programs that have always been a focus. This paper first describes the goals and history of the program, and follows with a discussion of SRC programs. With consideration of important management aspects, we assess which characteristics of the organization affect its success, consider these characteristics in regard to other innovation programs, and address the question of whether the lessons we learn from SRC are diffusible to energy innovation institutions.

Methods

We gathered data from this case study in semi-structured, qualitative interviews, conducted between March and December of 2010 (Bozeman 2000). Interviewees at SRC included the organization's president, program leaders, and other decision makers. 19 people participated in the interviews, including 9 staff at the institution, 7 representatives from industry members, and 8 current and former SRC grantees, with several having served multiple roles. Most interviews were 30 minutes to 1 hour in length, with several interviewees participating in multiple interviews. Interviews explored what decision processes look like, which factors play a role in decision making, who participates in decisions, and what constraints exist. Additionally, following the idea that salience and credibility in research results can stem from relationships with user communities (Cash et al. 2002), interviews asked how SRC targets users, which mechanisms enable user satisfaction, and how users evaluate the institution.

In addition to interviews, much of the data came from an examination of SRC budget figures and various internal documents. This information provided both the specifics on institutional operations and fine detail on how SRC has changed over time. Such data illuminated SRC's evolving relationship with member companies, how its program structures have changed to accommodate such relationships, and evolving relationships between SRC and the government.

SRC and its Mission Goals

The original SRC members included IBM, National Semiconductor, Advanced Micro Devices, Intel, Control Data Corporation, Digital Equipment Corporation, and AT&T Bell Laboratories, with the number of participants soon growing to over a dozen, and including companies such as Texas Instruments and Motorola. As the problem definition changed, from one where international competition in semiconductor technology was at the forefront, to one that emphasizes building a healthy semiconductor industry as a whole, the SRC has become international in membership. The mission of SRC is specific and clear: it "...is to manage a range of consortial, university research programs some of which are worldwide, each matching the needs of their sponsoring entities. SRC maximizes synergy between programs to optimally address members' research needs and minimizes redundancy

to maximize value to common members.”² SRC works toward these goals by managing the boundary between the industry members and university academics. SRC reinforces its mission with abundant involvement of industry members. Two central aspects to mission achievement are graduate training in semiconductor research and continuing semiconductor innovation for the medium- and long-term.

SRC programs each have separate missions, each related to the institution’s overarching goals. For example, GRC aims “to provide for innovative, strategic, pre-competitive research guided by the ITRS [International Technology Roadmap for Semiconductors], conducted in universities worldwide,” and to provide “a global forum for pre-competitive collaboration among all segments of the semiconductor industry.”³ FCRP also funds universities, but for “long-term breakthrough research that results in paradigm shifts and multiple technology options.”⁴ NRI looks to even longer term research, focusing on novel computing devices capable of replacing the CMOS transistor as a logic switch in the 2020 time frame. ERI, focusing on the energy sector, has a mission to “enable reliable, low-cost renewable energy systems and efficient energy use and distribution through enabled and optimized smart grid.”⁵ All programs promise a specific kind of research, bounded by time frame or sector, and do so with an aim toward application for industry members. They also all focus industry dollars to precompetitive research, which might provide the foundation for further profits from discovery. By leveraging the precompetitive stages, the companies are able to gain more, in the form of research results, but still compete against each other in terms of technological end products.

As an industry consortium, the institution is not subjected to the same management issues that institutions within the government face. For example, the funders are from the same organizations as the customers; such a relationship forces SRC to continually meet user needs to retain its funding. Government agencies can have a similar relationship with funders in Congress and the Executive Branch, when they are supplying information directly to these decision makers. However, they also have a predominance of users in the greater constituency to whom they are not always accountable. In SRC, member companies fund the work, serve on evaluation boards, and interact with the academics that perform the research. The system guarantees that a company can be actively involved in obtaining what it wants from the process. They have the ability to leave SRC if displeased with its direction and results, providing incentive for SRC to keep its members satisfied. To retain members, SRC has shown itself to be flexible in its policymaking, with the ability to alter processes to meet member companies’ needs evident in many of its decisions.

All of these characteristics point to SRC as an institution that first, has a mission that its member companies support, and second, continues to produce results that these companies believe are pertinent. Added to this clarity of mission is simplicity

² <http://www.src.org/about/mission/> accessed 07/11.

³ <http://www.src.org/program/grc/about/mission/> accessed 07/11.

⁴ <http://www.src.org/program/fcrp/about/mission/> Accessed on 7/11.

⁵ <http://www.src.org/program/eri/about/mission/> Accessed on 7/11.

in process that allows SRC to operate cleanly and efficiently. Within each program, member companies pay according to set rules, which in the case of GRC, FCRP, and NRI, are based in part on the size of the company in terms of sales. For the ERI, each member company pays the same amount. SRC distributes the money per the recommendations of advisory boards sent from the same industry members. Once projects are chosen based on their desirability to industry, both SRC and member companies interact with the researchers through annual reviews, conferences, and liaison interactions.

International Participation and Other Members

Despite worries of international competitions, SRC originally allowed participation by foreign companies, with the provision that they base membership fees on worldwide sales (as did all participants), and extend intellectual property rights within the home country that were identical to the non-exclusive, royalty-free licenses extended within the U.S. Additionally, U.S. companies wanted to be able to participate in cooperative research programs in the home country of any member. After a year without foreign participation, SRC altered its membership rules to only include U.S. companies, both to leverage more federal government money, and to increase the competitive advantage of U.S. companies. Until 1999, when these were seen as less of an issue, SRC participation was limited to U.S. companies. Currently, two programs are international. However, the two current SRC programs that leverage the highest proportion of U.S. funding are still limited to U.S.-based companies. In SRC programs with government partnerships present, the federal agenda does not significantly drive the prioritization and implementation process, beyond broad guidance. While federal partners do play a role in advising and maintaining relationships with SRC and its members, they do not have an overt amount of influence on the process.

SEMATECH has also supported university research through SRC. SEMATECH, which stands for semiconductor manufacturing technology, is related to SRC in that it also formed as a collaborative group for solving problems in the semiconductor industry, usually focusing on manufacturing problems, and has had many shared members, like IBM and Texas Instruments. Like SRC, it is a collaborative consortium, started by member dues from chip manufacturers and other interested parties. However, unlike SRC, SEMATECH itself focuses mainly on manufacturing processes rather than semiconductor technologies. SEMATECH has also been lauded by the Department of Energy (DOE) as a model on which to base solar energy technology development, and is involved with the DOE's \$110 million SunShot program to enhance such technologies. SEMATECH's involvement with SRC goes back to the 1980s. A small amount of SEMATECH funding has gone to SRC. From the beginning of SRC to 1996, SEMATECH had provided 22% of SRC revenue, mostly by funding SRC to support its university led SEMATECH Centers of Excellence.

Where SRC Succeeds

Institutional Structures and SRC Success

Burger (2000) provides a list of things that might not exist without SRC. They include:

1. a relevant, productive, silicon device related, university research program in U.S. universities,
2. IC [integrated circuit] research in design, microstructures, manufacturing, and packaging sciences currently being used by U.S. industry,
3. an additional 3,000 graduates [nearly 9,000 in 2011] with advanced degrees and relevant IC research experience added to the industry's manpower base, with an additional 300 graduating every year
4. industry technology leaders working together to define needs and strategies through the 'SIA Technology Roadmap for Semiconductors',
5. hundreds of industry scientists and engineers serving as mentors to university faculty and students engaged in SRC's semiconductor research,
6. the National Advisory Committee on Semiconductor and its successor, the Semiconductor Technology Council,
7. significant additional support for integrated circuit research that was incentivized by the SRC,
8. SEMATECH, and
9. an altered paradigm for cooperative R&D.

In 2005, SRC won the National Medal of Technology:

"For building the world's largest and most successful university research force to support the rapid growth and advance of the semiconductor industry; for proving the concept of collaborative research as the first high-tech research consortium; and for creating the concept and methodology that evolved into the International Technology Roadmap for Semiconductors."⁶

The continued participation of the Department of Defense's (DOD) Defense Advanced Research Project Agency (DARPA) is also evidence of a perceived value from SRC research. According to the SRC director of the FCRP, DARPA buys in to SRC's general approach, and to the general significance of pursuing long-term investments in the technological infrastructure. Specifically, DARPA may see semiconductor advancements as key for military advancements. For example, new systems for updated fighter jets depend, at the base level, on semiconductor improvements, as do many other computing and information systems.

SRC is not useful for every semiconductor corporation; many in the field do not buy in. Those that abstain presumably do not believe that the benefits outweigh the costs in funding and personnel time. As the industry becomes more fragmented, many companies do limited or no manufacturing and are only interested in SRC's

⁶ U.S. Patent and Trademark Office: <http://www.uspto.gov/about/nmti/recipients/2005.jsp> accessed 11/10.

design-related research. Many start-ups do not join because of their relatively small size. While such companies might also gain from working together in a consortium, and while they might also see workforce benefits from student development, they do not see these as worth the price of admission due to their typically smaller budgets and limited ability to commit human resources. The model does lend itself to participation by larger, vertically integrated companies. Given the growing scope and cost of research needs to remain on pace with Moore's Law, SRC members continue to see value in leveraging their investments in the earliest stage, precompetitive research. Most of the SRC members do not have the size and research capabilities to carry out long-term research on the scale of a Bell Labs, but they do have the resources to both pay into research, and to harness that research to develop new technologies and put them on the market.

Outputs from SRC-Funded Research

In terms of results, SRC has been successful in several ways. First, the research has helped to push technology forward to keep pace with Moore's Law and contribute to other developments that will move innovation forward. While this first does not solely aid the member companies, it does help them meet emerging market needs. The National Medal of Technology credits SRC with supporting growth in the industry.

Publications and citations are not a direct measure of innovation or effectiveness, but SRC does use them to estimate research output and impact. Many thousands of peer-reviewed publications are based on SRC-funded research. While a complete list is not available, SRC has record of more than 125 publications that have received at least 100 citations. Of those, the number of papers for which more than 15% of citations are by industry—an indication of potential commercial relevance—is 72 (or 58% of the total).

Patents represent research outputs that are more closely tied to innovation and potential economic benefit. Since its inception, 377 patents have been granted based on SRC-funded research.

Measuring number of publications and patents is far from the only way success in technology can be estimated, and it neglects other benefits, like workforce development, capacity building, strategic orientations, and tacit knowledge transferred through interactions between researchers and member companies.

SRC-funded research has contributed to several important developments in the field. For example, SRC began work in lead free flip chip packaging, which is a means for connecting chips to external circuits, in the 1990s. One of the most influential papers in the area (Laird 2001) emerged from SRC funding in 2002, and now had 578 citations, 20% of which are from industry. Additionally, SRC supported nine other publications with over 100 citations in this area. Commercialization on the products began in the mid-2000s. They are utilized by several SRC members, and have the capability to address the European Union's environmental standards.

Several SRC programs have led to important innovations. One of these, a tri-sided metal gate FINFET, will allow for increased speed and lower power consumption in future microprocessors. SRC has played a role in every major component of such devices, which are being commercialized by Intel. Such a development is important because it has extended the life of current manufacturing techniques. Since the amount of sunken cost in existing manufacturing infrastructure is high, this allows for advancement of transistors with less investment than a new technology with radically new manufacturing processes might require. By funding work in the late 1990s, at the University of California-Berkeley and elsewhere, SRC was able to push this innovation forward.

Workforce Development

Beyond funding useful research, SRC has worked to meet members' needs through workforce development. In the early 1980s, one main concern of the industry was a lack of skilled workers in the burgeoning areas of research for the industry. By funding students to work in areas where they would develop the necessary skill set, SRC has expanded the talent pool for the industry. By hiring graduating SRC students to work at their companies, industry members enable a continued job market demand for the students SRC funds. An additional example exists in SRC participation in the ITRS. This falls outside the responsibility for funding research, but allows the institution to repeatedly inform, and be informed in, emerging trends in semiconductor research needs. According to SRC, "One of the fundamental successes of the SRC model ... has been to help prepare graduate students for effective roles in semiconductor industry." In 1982, the graduate students who participated in industry-relevant research numbered less than 100. These numbers have increased greatly, due to SRC funding (Table 1).

As a whole, SRC supports almost 1,500 doctoral students annually, preparing them for positions in the semiconductor industry. SRC approaches development of future workforce as a singular part of its mission. A large portion of SRC funding has gone to pay for graduate students in relevant fields, to encourage expertise in the areas where SRC members need them. For example, one of the things ERI seeks to develop is "relevantly educated graduates with the expertise and skills to transition ... new methods into industry" and "A research center devoted to the development of conceptual approaches and computational tools for energy conversion devices would provide companies with graduates possessing skills needed for this emerging industry" (Fuchs and Erica 2009).

The use of industry funding to produce a generation of young people expert in the semiconductor field is also integral to the success of SRC. SRC has built a base of graduate students to comprise its talent base. The number of U.S. dissertations with the descriptor 'silicon' almost tripled, from 162 to 470, between 1982 and 1990. In the same time period, dissertations using the key word 'integrated circuits' went from 31 to 110 (Etzkowitz 1984). Many of these dissertations emerged from SRC funding. These students contribute valuable semiconductor research during their

Table 1 SRC Graduate students funded

Program	2011 Students	Students to date
GRC	726	6,861
FCRP	520	1,963
NRI	210	382
ERI	22	23

graduate careers, and many go on to work in the semiconductor industry, ensuring continued innovation.

For many of the graduate student grantees of SRC, a job in the semiconductor sector, often working for an SRC member, is a common next step. Some funding is contingent upon students taking a job with an SRC member following graduate study. Many of the interviewees who were previous graduate fellows of SRC related that SRC interest in a topic area had altered their direction of inquiry for dissertation research, typically not drastically, but in a way that guided output towards SRC, and member needs. Many spoke of the processes that enabled interaction, like annual meetings with SRC that enabled two-way communication about projects and outputs, and informed graduate students of future job opportunities in the field. This type of process also enables students, and their academic advisors, to be kept aware of emerging needs of the semiconductor industry, allowing for short-term workforce development by increasing the salience of current student work, and informing future processes of student and project choice by the professors, resulting in longer-term relevancy gains.

Evolution of SRC Decision Processes

SRC Programs

SRC used to consist of a single program, but it has added several more over the years. Each program has its own set of member companies that buy into the mission of that program. While many companies buy into more than one program, this is contingent on the corporations' interest and willingness to participate in each program individually. Each program has a larger focus, and is then divided into a number of topical component areas.

Within GRC, all funds from companies used to go into one common pool; SRC has changed policy in recent years to allow companies to direct a portion of their funding to projects that they believe are promising. This allows members to continue leveraging the larger pool of funding that SRC provides, since they are not the only group contributing to a particular research thrust. They are directing a portion of their funds to buttress further a research area that they believe will be promising.

SRC's first program was devoted to research addressing nearer- and medium-term needs. With the success of these programs, the organization has grown to support fundamental research more recently. The Global Research Collaboration (GRC) is the part of SRC that encompasses the original program. The current

members of GRC include Advanced Micro Devices (AMD), Freescale Semiconductor, GlobalFoundries, IBM, Intel, and Texas Instruments. AMD, IBM, and Intel were all original members of SRC. GlobalFoundries and Freescale were spun out from AMD and Motorola, respectively, so are also tied to founding members. GRC exists to fund relevant research that can extend the scaling, and thus the capabilities, of complementary metal-oxide semiconductors (CMOS), the prevalent technology for constructing integrated circuits. The purpose is keeping stride with Moore's Law, which describes the trend of doubling the number of transistors on integrated circuits every two years, and thus increasing attributes like memory and processing speed. GRC typically operates through three-year contracts between SRC and the universities, and concentrates in areas where the output is expected in a 3–5 year span. In addition to the money that industry contributes, GRC leverages significant funding, often from government sources, to contribute to the research. Between 1997 and 2010 the amount of leveraged funding, including collaborative funding, SRC-directed money, and SRC-influenced funding varied between 8% and 73% of total funding; it's been over 60% for the last 6 years.⁷ The overhead costs for GRC are typically 12% to 13% of the funding provided by industry, which they cite as an extremely low cost to pay for the service of managing research programs and leveraging much more funding than one member can provide.

Since the member companies ultimately manage GRC, the Board of Directors consists of representatives from the companies and has overall responsibility for GRC's direction. Additionally, an Executive Technical Advisory Board (ETAB) provides broad technical guidance to the program. In the five science areas within GRC, a Science Advisory Coordinating Committee participates, and a Technical Advisory Committee helps to direct each thrust. Science areas include Device Sciences, Computer-Aided Design and Test Sciences, Integrated Circuits and Systems, Nanomanufacturing Sciences, and Interconnect & Packaging Sciences, and each science area has between two and five thrusts.

Inclined to longer-term results, the Focus Center Research Program (FCRP) investigates technologies by funding interdisciplinary centers (currently there are six) that integrate multiuniversity teams for solving problems. FCRP members include Applied Materials, GLOBALFOUNDRIES, IBM, Intel, Micron Technology, Novellus Systems, Raytheon, Texas Instruments, United Technologies, and Xilinx. Within each center, a Director at one of the participating universities leads the effort. Member companies provide funding for FCRP in partnership with the Defense Advanced Research Projects Agency (DARPA), which contributes about half of the funding.

Instead of focusing on Moore's Law, NRI pursues next generation technologies (Etzkowitz and Spivack 2001). The National Institute of Standards & Technology participates in funding several NRI centers with industry members consisting of GLOBALFOUNDRIES, IBM, Intel, Micron Technology, and Texas Instruments. The governments of four states and one municipality participate as strategic

⁷ The types of leveraged funding are: "directed"—funds from a non-industry sponsor are transferred to and managed by SRC; "collaborative"—funding by a third party, usually a government funding agency, that is invested jointly with SRC but that is transferred directly to the university; and "influenced"—funding by a third party that is invested in SRC-relevant research based on input from SRC.

partners. NRI also funds several joint projects with the National Science Foundation (NSF), leveraging the NSF's network of nanoscience and materials research centers.

SRC instituted its most recent program, Energy Research Initiative (ERI), to leverage industry member funding in precompetitive alternative energy technologies. The mission states that it will “enable reliable, low-cost renewable energy systems and efficient energy use and distribution through an enabled and optimized smart grid.”⁸ Rather than focusing on timescale or type of research, ERI currently focuses in two places, Smart Grid technology and photovoltaics. Over time, SRC hopes to grow ERI so that it will consist of a larger number of centers, each with multiple universities participating. Members of ERI include some companies that participate in other SRC programs, like IBM, Applied Materials, and Tokyo Electron, and some that are closely tied to the energy industry, such as thin film manufacturer First Solar, power technology company ABB, cabling system manufacturer Nexans, and Robert Bosch GmbH.

ERI seeks to use strategies that have been successful for semiconductors, and to transfer similar strategies to energy research. Currently the two programs are centered at Purdue University (PV) and Carnegie Mellon (Smart Grid). While photovoltaic technologies include semiconductor technology in many solar cells, PV systems do differ in that there is a unique set of technical issues, including storage and transmission, surrounding the technology. Furthermore, the range of application is more limited than for semiconductors, which have been utilized in a much wider context than that of solar energy production. Additionally, as early as the 1970s, solar technologies have experienced major policy hurdles (Laird 2001) and suppressed growth due to lack of support and subsidy (Etzkowitz 1984). However, rather than transfer its technological expertise, SRC is looking to apply its knowledge on managing science. In electronics, “the pervasive use of simulation in semiconductor process development, device design, and system analysis” enhanced the success of the products. SRC argues that development of similar aspects of solar technology could lead to great benefit (Fuchs and Erica 2009).

Table 2 compares SRC's programs in seven categories. Importantly, each program is different in the way it contributes to SRC's greater mission. Considering the first three programs (GRC, FCRP, and NRI), one can see that the first focuses on finding an appropriate pathway to take the current dominant technology, CMOS; the second attempts to expand on technological options to assess next pathways in semiconductor technology, and the third looks to explore the foundations of next generation (i.e. post-CMOS) devices. Such a set of offerings, with sponsors able to join any one program, or all of them, allows members to focus on the medium- or long-term, as they see appropriate to their needs and opportunities. Additionally, involvement of the industry members, and the autonomy of the researchers, varies with how close the program is to concrete applications, obtaining more input for nearer-term products, to ensure a valuable discovery. With this, the mechanism for delivering products to industry members varies as well. Such a system caters to different industry members' needs based on the pertinence of the problem or

⁸ <http://www.src.org/program/eri/about/mission/>.

Table 2 SRC Research Programs (adapted from SRC figure)

	Global Research Collaboration (GRC)	Focus Center Research Program (FCRP)	Nanoelectronics Research Initiative (NRI)	Energy Research Initiative (ERI)
Nature of Research	Narrowing options-individual researchers and centers	Creating options-multi-university centers	Discover next-logic switch - individual researchers (at NSF centers) and multi-university centers	Approaches to electrical energy generation, storage, distribution, and infrastructure
Technology Spectrum	CMOS ^a	CMOS, "hooks" to beyond CMOS (Fuchs and Erica 2009)	Beyond CMOS (Fuchs and Erica 2009)	Enabling technologies in PV and smart grid
Industry Involvement	Strategic and in execution	High level strategic	Influence through highly leveraged co-investment	Strategic and in execution
Government Involvement	Limited; leverage funding	50:50 with DoD ^b ; program direction	Major funding - state and federal (NSF and NIST)	None, but SRC is open to the possibility
University Autonomy	Collaborative (Industry/university)	Significant	Significant	Collaborative (Industry/university)
Membership	Global	U.S. only	U.S. only	Global
Technology Transfer	Industry-driven processes (e.g. Industrial liaisons e-workshops, etc)	Primarily faculty-driven processes (e.g. periodic reviews, seminars, etc.)	Combination of industry & faculty driven processes (e.g. Industrial assignees, e-workshops, etc.)	Industry-driven processes (e.g. Industrial liaisons e-workshops, etc)

^a Complementary metal oxide semiconductors^b Department of Defense

discovery, the amount of time before industry uptake, and the ability of the technology to work within the current technological regime.

Acquiring Intelligence

Much of the process at SRC involves interaction with the industry members, allowing the members' needs to steer the decisions made at SRC and the research performed at universities. While universities make proposals to SRC for project funding, industry members define the strategic goals of SRC research programs, and SRC makes requests for proposals based on these parameters, and participates in research execution by including its own liaisons. Additionally, there are other mechanisms in place for SRC to acquire intelligence, about what members want both in the long and the short-term. First, the semiconductor industry, with involvement of SRC and its member companies, periodically updates the International Technical Roadmap for Semiconductors [ITRS]. Through an international membership, the ITRS looks "to ensure cost-effective advancements in the performance of the integrated circuit and the products that employ such devices, thereby continuing the health and success of this industry."⁹ The U.S., through the SIA and its member companies, contributes 55% of ITRS participants in a process that works to comprehend "the spectrum of needs for basic research capabilities and product potentials." Such an effort enables SRC to have an up-to-date, comprehensive view of where the semiconductor industry believes it is heading, and of its goals for future technological attainments.

SRC, through its chief scientist, also holds small forums that are intended both to challenge the participants' thinking, and to help maintain an awareness of the status quo, but also to plant new seeds for SRC research, by serving as a venue for information sharing. Such forums invite leaders from the field to discuss SRC-related issues and to present their new work. They are typically open only to those invited, and include joint participation between university and industry representatives.

Additionally, the direct processes through which SRC members participate in its decision processes allow for the organization to be aware of needs on a more operational, short-term basis. Such processes are discussed in the following section.

Leadership and Advice

SRC's leadership structure is fairly simple, allowing for efficient management within the corporation along with participation by industry members through a number of advisory boards. Many of the leaders of SRC have been involved with the institution for a long time, and its president, Larry Sumney, has been running the institutions since its inception in 1982. This 30-year consistency in the face of changing market dynamics and industry needs speaks to the ability of SRC leadership to continue delivering results to its members. Much of the role of SRC leadership is to serve as a liaison between industry and academic researchers. As

⁹ <http://www.itrs.net/about.html> last accessed on 07/11.

such, a large role is played by the institution's advisory boards. These advisory boards consist of representatives from the member companies, and direct both the short-term technical direction and the strategic planning. Interactions during the prioritization process and during research itself drive SRC. According to one author (Burger 2000), "Inputs from SRC member companies through the Board of Directors, the Technical Advisory Board [TAB] and the mentors assures the continued responsiveness of SRC research to industry technology needs. The challenge is to distill from the variety of inputs, a coherent agenda that is consistent with the needs of a majority of the membership."

For GRC, one of the ways that industry advisors have inputs is by rating projects that the program managers are considering or conducting. Members can rate projects using an IS (Importance and Satisfaction) score, which stands for a number that is the product of ratings for both the project's importance (in meeting current technology goals) and satisfaction (of the scoring company) with the progress being made. By integrating such scores into the decision process, SRC can alter its path towards those ideas that users feel are more likely to contribute benefit, and can thus be flexible to keep their members satisfied.

The advisory board member composition changes with the level at which it operates. The Board of Directors and the Executive TAB might include high level company decision makers, such as Vice Presidents of Research or Chief Technology Officers, to input higher order, strategic decisions made about SRC direction. The advisory boards that interact at the level of strategic areas and science thrusts might contain a higher number of technical program managers, who can contribute more knowledge and needs awareness on specific issues of technology direction and usefulness. This structure allows for a multi-tiered interaction between company experts, SRC employees, and academic researchers, to guide the process both towards strategic usefulness and short term technical need.

In addition to broad guidance by the advisory boards, many of the projects in SRC programs get guidance from yearly program reviews, these reviews serve as a means for researchers to disseminate their results within SRC, and for members to critique the work and steer the direction for the next year. While projects usually fulfill their three-year terms, SRC programs can use annual reviews to guide and shape them as they develop.

A more personal, iterative relationship is also encouraged by the existence of industrial liaisons. Liaisons, who are employees from the members, work with the university researchers to provide input on a project so that it has better chances of meeting an industry need and eventually being commercialized. By making regular personal contact with the researchers, liaisons can help both the mentoring process for graduate students and the likelihood of a usable outcome.

Finally, SRC operates TECHCON, a research conference and job fair that allows students working on SRC research to present to, and interact with, people in the industries that are supporting them. According to interviewees who were past SRC-funded graduate students, TECHCON was both a useful venue for learning about the state of the research and industry, and for securing the jobs they worked in following their studentships.

Industry and Federal Support

Several large companies have consistently funded SRC's component programs for almost 30 years (Fig. 1). The industry funding for SRC, across all programs, has gradually risen. The small downturn in numbers in 2009 and 2010 represents reduced membership fees due to the global downturn. Since member payments are based on a percentage of sales, the fees were lower during the downturn. Not that, even across all of its programs, the size of the SRC budget is modest, in the area of \$150 million, it has half the funding of a lab like the National Renewable Energy Laboratory, and less than one tenth the funding of a larger national laboratory like the Sandia National Laboratory or of Bell Labs at its peak. The willingness of companies such as IBM to stay involved with SRC programs is evidence that the organization has been consistently meeting the expectations that its member companies set for it. The amount of funding from government sources, represented in the "Other" category of Figure 1, has also increased dramatically, from around 8% in 1997 to over 70% in 2010, representing a consistent increase in government contributions, which has two components. First, the addition of programs with longer time horizons, such as FCRP and NRI (started in 1998 and 2005, respectively), has brought in increased government involvement compared to the nearer-term GRC. And second, the other category includes NSF funds that are not managed as part of SRC's research investment, but that support university research that aligns with SRC research program and goals. NSF-supported researchers may be invited to participate in SRC reviews and provide research results to SRC for sharing with member companies. However, their research is not directed by SRC.

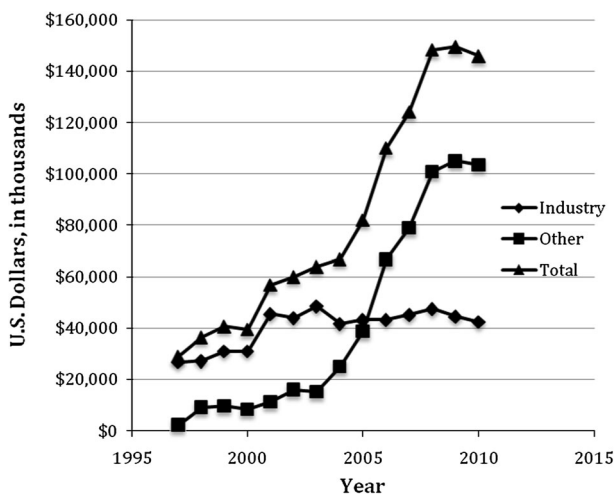


Fig. 1 SRC Funding to universities across GRC, FCRP and NRI, including industry member fees, other funds, and total funds. Other funds include a small amount of funding from SEMATECH, but mainly consist of government funding, some of which is directed by SRC and some of which is directed by the government agency

In contributing a large portion of funding for FCRP, DARPA is endorsing the SRC model as an effective one. DOD has participated in FCRP from the beginning; industry members conceptualized the program as a collaboration with the Defense Director of Research and Engineering (DDRE), with a Memorandum of Understanding signed by Intel and DDRE. SRC was brought in at the start to manage the program. Originally, DDRE and DARPA both contributed funding to FCRP; DARPA has since taken over as sole DOD funder, providing a growing amount, of \$17–20 million per year over the last five years. As with any other member, the DARPA contribution to the program is on a yearly basis, and they do have the option to quit. However, as of 2008, DARPA has included the FCRP funding as a discreet line item in the budget.

Although it contributes more to the budget than any other FCRP member, DARPA plays the same role in shaping and prioritizing research as any other. A one vote per member rule holds for DARPA, just as it does for the member companies, which differ in size of contribution amongst them. As a member, DARPA participates in both high level decision-making and science advisory boards, providing guidance and input to the SRC process.

SRC programs consist of work carried out by academics and the research is open to publication and academic dissemination. However, the SRC's goal is to allow its member companies early and unrestricted access to SRC funded advancements. It accomplishes this by providing near real-time access to results via a protected website and regular e-seminars and in-person reviews, and by making all intellectual property fall under non-exclusive royalty free licensing, meaning that all members can use the innovations without paying additional funds beyond membership fees. Thus, the advantages of innovation are shared by all of the members paying into the particular program, whether it is GRC, FCRP, NRI, or ERI, that develops it.

SRC funds have overwhelmingly gone to research. The SRC has kept its level of membership fees devoted to overhead at or below 13%. This number is impressively low, and allows most of the money to pass through SRC to researchers. While there is a second toll of university overhead on the funding, the SRC burden on the member universities is low. The result of an industry member's contribution is that, although it's funding is reduced by 13%, the 87% that does go to research is greatly augmented by the contributions from other members and the government.

Second, SRC has continually educated graduate students in semiconductor science and engineering, thereby providing a workforce for its member companies. By ensuring a significant number of emerging PhD's with research skills in pertinent areas, SRC has worked to create a valuable commodity. Building human capital, both through the relationships created between industry and academic experts and by actually developing a workforce fluent in semiconductors, is an important part of technology transfer that has been often overlooked in trying to measure the effectiveness of technology activities (Logar 2009).

Finally, SRC has continuously contributed to road-mapping activities for the semiconductor industry. Semiconductor road-mapping has been a successful way to strategize the future direction for SRC and its companies specifically, but also for

the larger semiconductor area they operate within. SRC utilizes such roadmaps in deducing the direction of its thrusts and science areas.

Many aspects of SRC work have enabled it to succeed in its role as an organization tying academic research and industry need. Several of these things are similar to attributes of some other successful institutions. For example, DARPA, with a 50-year record of success in its field, has a lean and flexible management structure, is outcome focused, and has instituted a process that enables prioritization “as much from listening to the voices of active researchers, and nudging them in common directions, as they do from listening to their . . . customers” (PCAST 2010). The DARPA structure quickly toward making funding decisions; program managers obtain input from experts in the field, and have the leeway to make the decision from that input. It may follow a different operational structure, but DARPA, like SRC, utilizes the views of their technology consumers in their decision processes. DOE is attempting to emulate this lean administrative structure with the creation of the Advanced Research Project Agency for Energy (ARPA-E), in an effort to bring similar successes to energy innovation. Such an effort could move money to potentially useful areas without the burdens associated with operating in administration-heavy offices, such as Office of Science, and without relying on the available resources of on national laboratories.

The structure of the organization, where those funding the organization are also the chief users of SRC products, has kept SRC’s concerns closely tied to member concerns. Members involve themselves with multiple aspects of the decision process, from intelligence gathering to evaluation, and at multiple levels, including liaisons for individual projects and advisory board oversight at the highest levels. This involvement in multiple aspects of the SRC process means that, while SRC is managing the boundary between industry and the academy, the funding members are helping to manage the boundaries both between SRC and members, and between SRC and academia. In other words, they are taking an active role in managing their own boundaries.

In addition to the involvement of members, SRC has devised a successful means for their participation. The implementation of an IS score allows for the creation of a numeric metric through which the preferences of different participants can be compared. At the same time, while such scores allows participants to consider both the impact and the members’ level of satisfaction of a research problem, they allow the members to make their own judgments on how important, and how risky, such projects are. Thus, member preferences can inform the process, and can be compared to each other on an easily relatable basis. Additionally, the industry liaisons enable more in-depth participation, with members both informing processes and informing the implementation of research. The expectation would be that industry members only have incentive to pay for liaisons’ time in cases where the projects are of particular interest to them, again allowing them to shape the areas of SRC research that they feel could have the highest impact on their own operations.

The authors conducted a recent study on NREL that considered its organization and management, and conditions by which it might be successful (Logar, Anadon, and Narayanamurti, in review). The research suggested three areas in which innovation institutions like NREL and SRC might strategize to increase their

effectiveness. These areas are: 1) Contextualize institutional knowledge within the application area, 2) Increase social capital and legitimacy by building relationships with users and information consumers, and 3) Institute methods for both consistency and flexibility. The SRC mechanisms that have encouraged its successes have fallen within these categories. For example, road-mapping activities and liaison relationship allow for contextualization, and, along with advisory board processes and hiring of former students, increase the social capital built within the participants. Additionally, workforce development builds consistency in the availability of a strong talent pool, and SRC's willingness to alter its funding model, and to allow participants in GRC to allocate money to preferred program thrusts, speaks to the willingness of the organization to be flexible in the face of changing relationships. Such processes strengthen SRC in its field, and speak to the existence of transferable lessons for other institutions to pick up and develop upon.

SRC Challenges

This paper assesses the SRC as largely a successful organization, in terms of meeting its goals and maintaining industry support. However, there are challenges and shortcomings to the model. As previously mentioned, the model is one that does not encourage the participation of smaller companies. SRC benefits, in terms of workforce and innovation, may spill over to benefit smaller groups, but the opportunity to participate in shaping the SRC agenda, and to pool resources for common benefit is not attractive to smaller companies given SRC's membership structure. Thus, SRC could potentially provide shallow benefit to the whole industry, but in terms of deeper benefit, a rich-get-richer scenario is possible, where the large companies with the resources and desire to buy-in reap the rewards. Of course, a case may be made for a converse scenario, where larger companies are bearing the costs for activities that benefit non-members as well. In either case, finding a way to make participation more attractive to smaller companies is a missed opportunity by SRC.

Second, while largely stable, the budget is subject to the vagaries of the semiconductor market. For many SRC programs, member companies' dues are tied to sales. In times when sales are low, such as 2009 and 2010, the SRC budget suffers. This setup means that large drops in sales will drastically affect the SRC budget, with the result that the SRC will have a reduced ability to contribute money to innovation and workforce development, arguably during the times when the benefits of these activities are most necessary.

SRC and Other Institutional Models

The most surprising thing about SRC is that it has found a way to leverage industry money to motivate and develop a robust field of university research for over 30 years. SRC is successful in fulfilling the needs of its members, both public and private, and in delivering usable innovations. This model is one of many models that various innovation institutions have utilized. Another, for example, is that of single-

sponsor large industrial labs, which was successful for Bell. SRC connects research to application, and does so by supporting different programs, with different constituencies, to pursue both short-term advances in innovation and long-term fundamental research. Bell Labs, in its ability to fund long-term research while still marrying such research to both scientific and commercial accomplishments, succeeded by supporting the end-to-end system, from fundamental research to technology development, and linking successes in both to commercial development (Riordan and Hobbeson 1997). Such a system allowed for one institutional home to pursue all types of research with one institutional sponsor, harnessing more of the reward individually, but also taking on more risk for failed projects. Conversely, the SRC model allows different groups of sponsors to come together in one of the component programs, and to leverage their funding with inputs from other companies and federal sponsors, for a shared risk/reward system that can benefit the whole industry.

Just as different types of arrangements work for industries and different company sizes, we may need different types of models for various partnerships. SRC represents a good way for large- to medium-sized companies to fund research collaboratively. SRC had a “small company” membership category at one time, but the companies and SRC came to the conclusion that it was not worthwhile. Largely, smaller companies have not been included in SRC operations. Multiple factors may lead to this, including the short-term time horizon that smaller firms typically focus in, the lack of breadth in their research needs, inability to commit human and monetary resources to participation, and the lack of an existing infrastructure to utilize new discoveries outside of their specific expertise. Federal laboratories currently have capabilities for allowing small businesses to both participate in research activities and to utilize resources. Use of national laboratory user testing facilities, along with Work for Others arrangements, which allow companies to pay for laboratory services, can proceed with relatively small, one-time fees attached. NREL also has programs directed specifically towards small businesses, which allow for companies with fewer resources, or more specific needs, to be involved in strategic planning and research implementation. For example, NIST’s Nanofab, a fabrication facility for nanoscience research, allows smaller companies to utilize laboratory resources that would otherwise be inaccessible to them (Rubin and Rubin 1995). Conversely, SRC’s annual membership fee for GRC, with funds based on a proportion of sales, impedes the investment of small companies in their work. While the fee structure does allow for leveraging and stability, it does limit involvement.

Additionally, different technology products might lead to altered needs. Photovoltaic (PV) cells sit within a technological area that is similar to that of semiconductors, but are applied within a context that may put obstacles in the path from research to commercialization. PV cells have one application, to generate electricity, whereas semiconductors produce more differentiated products, in that one semiconductor technology can be used in many different consumer applications. This means that the precompetitive aspect of much semiconductor research might cover a broader span than in PV, where the manufacture of the cell is a large part of the sellable product. In selling electricity, the context for PV is also different, since it has to first, fit within the existing energy infrastructure, and second, replace

existing assets that have lifetimes of 40 years or more. Such a market means that the research stage of research, development, demonstration, and deployment is the simpler part of the task, and that the pathway to full-scale deployment might be outside the researcher's control. This does not mean the lessons from SRC are only applicable in its own case, but instead that such lessons may need to be flexibly applied for effective translation to other areas.

Applying Lessons from SRC to Energy Innovation Institutions

SRC has succeeded in several areas. They include:

- A stable but flexible budget, managed consistently for mission fulfillment and member needs.
- Effort extended to maintain member involvement in SRC processes.
- Examination of long-term trends and strategy-making to accommodate such trends.
- Integration of government interest, industry needs, and academic research participation.

SRC has a stably sourced, but flexibly managed, budget, along with mission and leadership focus that has aimed consistently at members' needs, and continuous involvement of the members in the advisory and research processes. The institution has continued to keep its members satisfied by training thousands of graduate students for employment in their companies, by focusing on precompetitive research that addresses commonly-derived goals, and by doing so in a manner that operates leanly, with low overhead to its funders. SRC can be a model for involvement of both large companies and federal funders in cooperative innovation. The institution does work in a market and context that is arguably simpler than that of many federal research institutions; the semiconductor industry does not always face the regulatory, policy, or market issues that energy does. Energy research occurs within areas where a highly uncertain future can obfuscate research choices and their outcomes. Although it may be difficult to utilize all of the decision-making flexibility and accountability to customers that SRC has in a federal institution, there are transferable lessons and strategies. For example, SRC has been successful at building and maintaining relationships with its industry partners, and the direct relationships between evaluators, funders, and industry members at SRC add effectiveness. Even in government-funded institutions, where ultimate responsibility lies in federal decision makers, more accountability to the users of research and development may aid them.

Buy-in by the user group is ensured at SRC, since the users are the funders/members and it would not be able to exist without member participation. In contrast, the relationship between federal laboratories and the named customer, industry, is not necessarily as tight due to the realities of federal funding structures. For example, the national laboratories could conceivably fail in their respective missions and continue receiving funding if appropriators are unaware of, or unconcerned with, such failures to adequately meet industry needs. For this reason,

building accountability to, and communication with, user groups is a necessity. Formal decision-making responsibility should always lie within the agency or the laboratory itself, since someone needs to be accountable for the institution's decision. However, an institution that allows inputs and guidance by including customers, in mechanisms that could be similar to those that SRC incorporates with its yearly reviews, could be successful. Advisory processes that formally include members of the industries served by laboratory research could enhance the incentives for the laboratories to work towards industry benefit.

SRC membership is based on multi-year commitments, therefore its funding is relatively stable. The situation is such that funding would only change drastically if multiple members were to leave or join the organization, but the multi-year commitments allow SRC to plan for such eventualities. The other major source of instability is the economy, which, by altering semiconductor sales, change fees by up to 30%. Federal institutions need to cope with the vagaries of budget politics. At the same time, DARPA has managed to consistently fund the FCRP for over a decade. Although defense funding may be less subject to political manipulation than energy innovation, there may be places in DOE's domain where common interest technology problems could be addressed by ARPA-E, or another group, in the same way DARPA has addressed those areas that the FCRP serves.

Even with an institution that succeeds in aiding its public constituents, Congress and the executive branch may cut the program for other reasons. Conversely, stability at SRC is likely as long as SRC keeps its members satisfied. However, the lesson to be learned is not that the SRC processes make stability more achievable; it is in how it does so, which is by tying its own metrics for success into its ability to serve its constituents. For many federal institutions, metrics for success are not always related to success in terms of outcomes. In research setting, peer-reviewed journal articles and similar outputs are always likely to be one significant aspect of the work. Energy institutions aiming at user applications must ensure that programs evaluate effectively based on other criteria, and pursue projects that will go on beyond peer-review publications to usable technologies. If academic outputs are the only output, and are not being tied to long-term success of the institutions' users, the metrics for success may need adjusting.

Additionally, SRC excels in its organizational flexibility. The administrative underpinnings of the institution are small, but it, along with the cooperation of its advisory boards and other member inputs, can change its research thrusts yearly to make new, more effective research policy when needed. To take an example from DOE, much of NREL's direction is managed outside of the laboratory, in EERE, DOE, and Congress. Such arrangements might make it difficult for NREL and other labs to adapt, with reduced ability to influence their program direction. NREL's current influence takes place through advising and giving recommendations to those making the decisions, placing more decision power, especially in the form of discretionary, lab-directed money, could aid the lab by first, making it responsive to upcoming problems or opportunities within its field, where it is both more expert and more able to respond quickly than the whole DOE could, and second, gaining responsibility and accountability for its decisions. Such an arrangement increases organizational flexibility, and makes considerations of needs more likely during decision-making.

Finally, the willingness of SRC and the semiconductor industry to convene and examine long-term direction, needs, and strategy has helped. A recent President's Council of Advisors on Science and Technology (PCAST) presentation stressed the importance of technology roadmaps in its policy proposals; SRC's long participation in ITRS road mapping is an important one. Examination of the road mapping process could lead to increased insight and more capability to consider energy projects with an eye towards innovation, energy technology roadmaps have been proposed by PCAST for use within the U.S. Department of Energy (SRC 2009). This kind of road mapping is one means for acquiring intelligence for strategic planning, but others exist, such as the process of soliciting white papers for program development, which can work to inform, and even adjust, research direction (Etzkowitz and Spivack 2001).

SRC has successfully integrated academic and industry concerns for years, and in some of its programs, like the FCRP and NRI, has included active participation of government entities as well. One potential outcome of the new ERI could be the development of valuable collaborative relationships between government, industry, and universities within the field of energy. SRC has been attempting to use its past successes in industry-university partnerships to promote new kinds of partnerships with federal laboratories that could be useful in addressing problems of energy innovation. The international membership of ERI may make it more difficult for a DOE group like ARPA-E to participate in ERI, but such membership could aid SRC in growing a program in a small, but developing, segment of its work to make significant contributions for innovation. SRC interviewees asserted that they would welcome a relationship with DOE, but one has not begun as of this writing.

As costs of research become large, cooperative funding is a useful model. While all of the specific mechanisms that SRC uses may not be transferable to energy innovation institutions, they can implement policies to maintain accountability to user groups, to focus more specifically on workforce development for their own, and their industries' future needs, and they can try to instill new mechanisms that are administratively lean, like ARPA-E, or to improve the administrative pathways in other programs. Although stable budgets are more difficult in the federal system, especially in politically charged fields like energy, they are something to aspire to when designing programs and funding new initiatives. Specific lessons from the SRC, like the use of liaisons between research users and researchers, could find useful application in federal research.

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Appendix: List of Interviewees

Larry Sumney, SRC President and CEO

Steven J. Hillenius, SRC Executive Vice-President

Celia Merzbacher, SRC Vice President, Innovative Partnerships

Elizabeth Weitzman, SRC Executive Vice-President and Executive Director, FCRP
 Jeffrey Walser, NRI Director
 Ralph Cavin, SRC Chief Scientist
 James Hutchby, SRC Senior Scientist
 Daniel Herr, SRC Director, Nanomanufacturing Science
 William Joyner, SRC Director, Computer Aided Design and Test
 Robert Esterveldt, Harvard University, Professor of Physics
 Charles Sodini, Massachusetts Institute of Technology, Professor of Electrical Engineering
 Walden Rhines - Mentor Graphics, Chief Executive Officer and Chairman of the Board
 Wilbert Van den Howk – Novellus, Former Executive Vice President and Technology Officer
 Allen Bowling, TI, Fellow and Manager of Research & Consortia, Analog Technology Development
 Sandy Aivolotis, Nexans, Senior Vice President, Operations, Technology and Business Development
 David Kyser, Applied Materials, Senior Director
 Derek Martin, Agilent Technologies, Digital ASIC Operating Manager
 David Newmark, Advanced Micro Designs, Design Fellow
 Paul Besser, GLOBALFOUNDRIES, Fellow

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